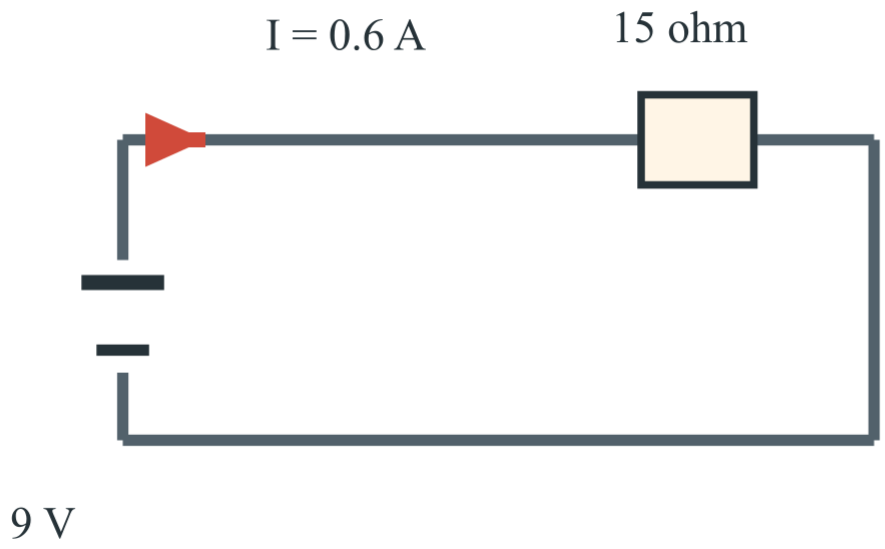
Battery-resistor DC circuit

This schematic shows a 9 V source and a 15 ohm resistor. The indicated conventional current is constant at `9 / 15 = 0.6 A`. The red arrow moves clockwise along the upper wire to indicate direction; its motion does not represent a changing current magnitude.

```
let supplyVoltage: Number = 9 let resistance: Number = 15 let current: Number =
supplyVoltage / resistance fn indicatorX(t: Number): Number = 58 + 25 * t diagram
batteryResistor {   title: "Battery-resistor DC circuit"   description: "A 9 volt
battery drives a 15 ohm resistor; the arrow travels clockwise to indicate
conventional current."   width: 320   height: 200   time t from 0 to 4 duration 4
line upperLeftWire {     x1: 50     y1: 130     x2: 188     y2: 130     stroke:
"#52616B"     lineWidth: 3     arrow: "none"   }   line upperRightWire {     x1:
218     y1: 130     x2: 250     y2: 130     stroke: "#52616B"     lineWidth: 3
arrow: "none"   }   line leftUpperWire {     x1: 50     y1: 130     x2: 50     y2:
98     stroke: "#52616B"     lineWidth: 3     arrow: "none"   }   line
leftLowerWire {     x1: 50     y1: 68     x2: 50     y2: 50     stroke: "#52616B"
lineWidth: 3     arrow: "none"   }   line rightWire {     x1: 250     y1: 130
x2: 250     y2: 50     stroke: "#52616B"     lineWidth: 3     arrow: "none"   }
line lowerWire {     x1: 250     y1: 50     x2: 50     y2: 50     stroke: "#52616B"
lineWidth: 3     arrow: "none"   }   line positivePlate {     x1: 39     y1: 92
x2: 61     y2: 92     stroke: "#263238"     lineWidth: 4     arrow: "none"   }
line negativePlate {     x1: 43     y1: 74     x2: 57     y2: 74     stroke:
"#263238"     lineWidth: 3     arrow: "none"   }   rectangle resistor {     x: 188
y: 118     width: 30     height: 24     stroke: "#263238"     fill: "#FFF5E6"
lineWidth: 2   }   line currentIndicator {     x1: indicatorX(t)     y1: 130
x2: indicatorX(t) + 14     y2: 130     stroke: "#D04A3A"     lineWidth: 4
arrow: "end"   }   text voltageLabel {     x: 20     y: 24     value: "9 V"
fill: "#263238"   }   text resistanceLabel {     x: 180     y: 156     value: "15
ohm"     fill: "#263238"   }   text currentLabel {     x: 88     y: 154     value:
"I = ${round(current, 2)} A"     fill: "#263238"   } }
```

Battery-resistor DC circuit

A 9 volt battery drives a 15 ohm resistor; the arrow travels clockwise to indicate conventional current.



The horizontal position of the arrow varies with diagram-local time. The current magnitude remains the computed value `0.6 A` throughout the displayed interval.